

Holographic warped CFTs from General Relativity on AdS_3

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ABSTRACT: We introduce a new set of boundary conditions for three-dimensional Einstein gravity with negative cosmological constant. The canonical generators plainly realize the semidirect sum of Virasoro and $u(1)$ Kac–Moody algebras in its standard form, without the need of additional redefinitions of the global charges. We show that requiring Euclidean BTZ and thermal AdS_3 to be diffeomorphic precisely induces the warped modular S –transformation of the chemical potentials. The corresponding warped Cardy formula then exactly reproduces the Bekenstein–Hawking entropy, where the ground state is identified as Euclidean AdS_3 . An explicit comparison with the boundary conditions of Brown–Henneaux and Compère–Song–Strominger is also addressed.

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1 Introduction

Warped conformal field theories (WCFTs) were introduced by Hofman and Strominger in [1]. Under similar reasonable and mild assumptions as in [2], in which global Poincaré and scaling symmetries are shown to enhance to the conformal algebra in 2D, they show that just requiring global invariance under translations in space and time and a chiral scaling implies that the symmetries are enhanced either to the full conformal algebra or to the semidirect sum of a Virasoro and a $u(1)$ Kac–Moody algebra. WCFTs correspond to the latter possibility. They are non-relativistic field theories, being neither Galilean nor Carrollian, since the generator of time translations behaves as a scalar instead of a vector under the $sl(2, \mathbb{R})$ subalgebra.

This class of field theories becomes of interest in different contexts, particularly for the microscopic behavior of near-extremal black holes. One of the best known examples of this sort is the near-horizon description of near extremal Kerr, whose $SL(2, \mathbb{R}) \times U(1)$ isometries correspond to the global ones of the warped conformal algebra [3]. Thus, WCFTs have spanned a growing activity along a variety of research lines [4, 5, 6, 7, 8, 9, 10, 11], including their relationship with entanglement entropy [12, 13, 14, 15, 16, 17, 18, 19, 20] and diverse holographic realizations along different setups [21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34].

Despite lacking full conformal invariance, this infinite-dimensional symmetry retains strong constraining power, leading in particular to warped modular covariance and a universal asymptotic density of states through the warped version of the Cardy formula [35].

The holographic realization of WCFTs is naturally formulated in three-spacetime dimensions, and it has followed three main routes¹: notably in the context of warped AdS_3 black holes in topologically massive gravity [35]; the so called “lower-spin gravity” [41], formulated as a Chern–Simons theory of $SL(2, \mathbb{R}) \times U(1)$; and General Relativity (GR) on AdS_3 with the boundary conditions of Compere, Song and Strominger (CSS) [42].

All of these holographic setups share a positive Virasoro central charge and a negative Kac–Moody level. Remarkably, as shown by Apolo and Song [43], the lack of unitarity displayed by descendants of negative norm is not actually fatal. Indeed, the modular bootstrap still remains viable provided the primaries are assumed to carry positive norm.

Here we propose a new set of boundary conditions for pure AdS_3 gravity that realize the semidirect sum of Virasoro and $u(1)$ Kac–Moody algebras directly in its standard form, with fixed central charge and level, given by

$$c = \frac{3\ell}{2G}, \quad k_{\text{KM}} = -k = -\frac{\ell}{4G}, \quad (1.1)$$

without the need of additional redefinitions of the canonical generators. The phase space includes BTZ black holes, so that in the Euclidean case, global AdS_3 becomes the ground state. One can then show that, requiring thermal AdS_3 to be diffeomorphic to Euclidean BTZ, successfully reproduces the modular properties of the chemical potentials under S -modular transformations, yielding to the warped Cardy formula [35], which in our case exactly reproduces the Bekenstein–Hawking entropy of BTZ black holes.

¹Other holographic approaches realizing the Virasoro–Kac–Moody algebra include [36, 37, 38, 39, 40]

Our results provide a minimal realization of warped holography in which the bulk dynamics is just described by pure GR, so that the structure of the dual WCFT₂ is determined by the new boundary conditions.

2 New boundary conditions for GR on AdS₃

In order to propose a set of asymptotic conditions being suitable to describe a holographic WCFT₂ we will follow the lines of [44, 45, 46] in which the fall-off of the metric is given in terms of a generic choice of lapse and shift functions. A very crucial point for our purposes is further allowing the lapse and shift to depend on the dynamical fields in a precise way at the boundary that is described in what follows.

For a localized distribution of matter, we assume that the leading terms of the asymptotic form of the metric are given by

$$ds^2 = g_{tt}dt^2 + 2g_{t\phi}dtd\phi - \frac{\ell^2\mu'dtd\rho}{\rho} + \frac{\ell^2d\rho^2}{\rho^2} + \rho^2 \left[1 + \frac{\ell^2}{4\rho^2} (\mathcal{P}^2 + 2\mathcal{P}') \right] \left[1 + \frac{\ell^2}{4\rho^2} (\mathcal{L} + \mathcal{P}^2) \right] d\phi^2, \quad (2.1)$$

with

$$g_{tt} = -\rho^2 \left[\mu - \frac{\mu_{\mathcal{L}}\ell^2}{4\rho^2} (\mathcal{L} + \mathcal{P}^2) \right] \left\{ \mu_{\mathcal{L}} - \frac{\ell^2}{4\rho^2} [\mu (\mathcal{P}^2 + 2\mathcal{P}') - 2\mu''] \right\} + \frac{\ell^2}{4} \mu'^2, \quad (2.2a)$$

$$2g_{t\phi} = (\mu - \mu_{\mathcal{L}}) \rho^2 + \frac{\ell^2}{2} [\mu (\mathcal{P}^2 + 2\mathcal{P}') - \mu'' - \mu_{\mathcal{L}} (\mathcal{L} + \mathcal{P}^2)] + \frac{\ell^4}{16\rho^2} (\mathcal{L} + \mathcal{P}^2) [(\mu - \mu_{\mathcal{L}}) (\mathcal{P}^2 + 2\mathcal{P}') - 2\mu''], \quad (2.2b)$$

where $\mathcal{L} = \mathcal{L}(t, \phi)$ and $\mathcal{P} = \mathcal{P}(t, \phi)$ stand for the dynamical fields, while the corresponding “chemical potentials” (sources), $\mu_{\mathcal{L}}$ and $\mu_{\mathcal{P}}$ are assumed to be fixed to a constant without variation at the boundary. Here $(\cdots)' = \partial_{\phi}(\cdots)$.

The asymptotic behavior of the metric then becomes fully specified once the remaining function μ is precisely defined. In order to carry out this task, as well as to further study the asymptotic structure, it is useful to introduce $\mathcal{I}_{\mathcal{P},\mathcal{K}}$ as the periodic solution of

$$(\partial_{\phi} - \mathcal{P}) \mathcal{I}_{\mathcal{P},\mathcal{K}} = \frac{1}{2} \mathcal{K}, \quad (2.3)$$

given by²

$$\mathcal{I}_{\mathcal{P},\mathcal{K}}(t, \phi) := \frac{1}{2} e^{\mathcal{F}(t, \phi)} \int^{\phi} d\varphi e^{-\mathcal{F}(t, \varphi)} \mathcal{K}(t, \varphi), \quad \mathcal{F}' = \mathcal{P}. \quad (2.4)$$

For the subsequent analysis, the relevant property of $\mathcal{I}_{\mathcal{P},\mathcal{K}}$ is that it satisfies the differential equation (2.3); while its explicit integral representation in (2.4) is not required.

²Periodicity amounts to discard the homogeneous solution of (2.3).

Thus, our boundary conditions become completely specified through the choice

$$\mu = -\mu_{\mathcal{L}} - \mu_{\mathcal{P}} \mathcal{I}_{\mathcal{P},1} . \quad (2.5)$$

The field equations in vacuum then imply that the subleading terms of the metric vanish, so that the family of spacetimes in (2.1) exactly solves the Einstein equations, provided the dynamical fields are chiral

$$(\partial_t + \mu_{\mathcal{L}} \partial_\phi) \mathcal{L} = (\partial_t + \mu_{\mathcal{L}} \partial_\phi) \mathcal{P} = 0 . \quad (2.6)$$

Note that $\mathcal{I}_{\mathcal{P},\mathcal{K}}$ carries this chirality when $\mathcal{K}$ also does, i.e., $(\partial_t + \mu_{\mathcal{L}} \partial_\phi) \mathcal{I}_{\mathcal{P},\mathcal{K}} = 0$. This holds in particular for $\mathcal{I}_{\mathcal{P},1}$, so that μ in (2.5) is also chiral.

Therefore, the boundary metric defined through $ds_{\rho \rightarrow \infty}^2 = \rho^2 ds_{\text{b}}^2$, which reads

$$ds_{\text{b}}^2 = (d\phi + \mu dt) (d\phi - \mu dt) , \quad (2.7)$$

becomes locally flat.

The BTZ black hole is recovered in the case of $\mathcal{L}$ and $\mathcal{P}$ constants³, provided that the following “geometric bounds”

$$\mathcal{L} + \mathcal{P}^2 \geq 0, \quad \mathcal{P}^2 \geq 0, \quad (2.8)$$

are fulfilled, which saturate at the extremal cases.

We should stress that global AdS_3 is excluded from the boundary conditions in Lorentzian signature. Indeed, in the case of $\mathcal{L}_{\text{AdS}} = 0$ and $\mathcal{P}_{\text{AdS}} = i$, the metric (2.1) becomes complex. Nevertheless, this configuration is allowed in the Euclidean case, since the metric becomes real and fits within our asymptotic conditions (see Fig. 1).

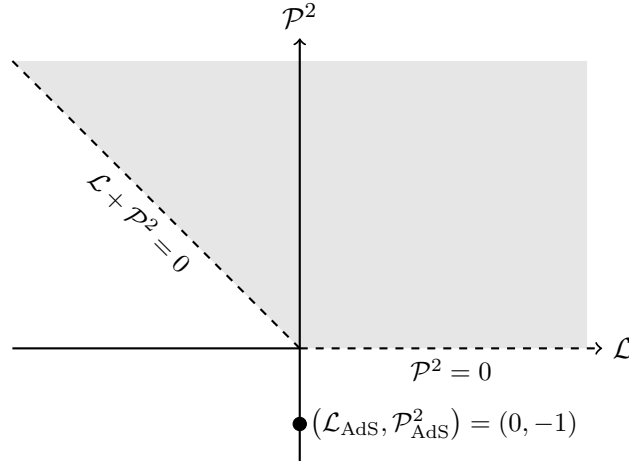


Figure 1. The BTZ black hole is defined in the grey region, satisfying the geometric bounds $\mathcal{L} + \mathcal{P}^2 \geq 0$ and $\mathcal{P}^2 \geq 0$, whose dashed boundaries correspond to the extremal cases. The black dot denotes Euclidean AdS_3 .

³Note that $\mathcal{P}$ can be chosen to be nonnegative ($\mathcal{P} \geq 0$), since the negative branch describes the same configurations but in a rotating frame at infinity.

2.1 Asymptotic symmetries and canonical generators of the WCFT algebra

The fall-off of the metric in (2.1) is preserved by asymptotic Killing vectors $\xi = \xi^\mu \partial_\mu$ that fulfill $\delta_\xi g_{\mu\nu} = -\mathcal{L}_\xi g_{\mu\nu}$, whose leading terms are given by

$$\begin{aligned}\xi^t &= \frac{\epsilon - \zeta}{\mu + \mu_{\mathcal{L}}} + \left[\frac{(\mu + \mu_{\mathcal{L}})(\epsilon - \zeta)'' - (\epsilon - \zeta)\mu''}{2(\mu + \mu_{\mathcal{L}})^2} \right] \frac{\ell^2}{\rho^2}, \\ \xi^\rho &= \left[\epsilon' + \zeta' + \frac{(\epsilon - \zeta)\mu'}{\mu + \mu_{\mathcal{L}}} \right] \frac{\rho}{2} + \frac{\ell^2 \mu' [(\mu + \mu_{\mathcal{L}})(\epsilon - \zeta)'' - \mu''(\epsilon - \zeta)]}{4(\mu + \mu_{\mathcal{L}})^2 \rho}, \\ \xi^\phi &= -\frac{\mu\epsilon + \mu_{\mathcal{L}}\zeta}{\mu + \mu_{\mathcal{L}}} + \frac{\ell^2 [(\mu + \mu_{\mathcal{L}})(\mu\zeta'' + \mu_{\mathcal{L}}\epsilon'') + (\epsilon - \zeta)\mu\mu'']}{2(\mu + \mu_{\mathcal{L}})^2 \rho^2},\end{aligned}\tag{2.9}$$

provided that the field equations (2.6) hold at infinity, and the dynamical fields transform as Virasoro–Kac–Moody currents, according to

$$\delta\mathcal{L} = (\mathcal{L}' + 2\mathcal{L}\partial_\phi - 2\partial_\phi^3)\epsilon - \eta'\mathcal{P}, \quad \delta\mathcal{P} = \frac{1}{2}\eta' + (\epsilon\mathcal{P})'. \tag{2.10}$$

Here, ϵ and η are field-independent chiral parameters fulfilling,

$$(\partial_t + \mu_{\mathcal{L}}\partial_\phi)\epsilon = (\partial_t + \mu_{\mathcal{L}}\partial_\phi)\eta = 0, \tag{2.11}$$

and ζ is given by

$$\zeta = \epsilon - \mathcal{I}_{\mathcal{P}, \eta+2\epsilon'}, \tag{2.12}$$

possessing the same chirality as in (2.11).

Note that the asymptotic Killing vectors manifestly depend on the dynamical field $\mathcal{P}$, which has to be carefully taken into account in order to define the canonical generators.

The variation of the corresponding surface integrals can be found in the Regge–Teitelboim approach [47]. They are independent of the radial coordinate, and given by

$$\delta Q = \frac{k}{8\pi} \int d\phi [\epsilon(\delta\mathcal{L} + 2\mathcal{P}\delta\mathcal{P}) + 2(\partial_\phi - \mathcal{P})\zeta\delta\mathcal{P}]. \tag{2.13}$$

Therefore, by virtue of (2.12) and (2.3), nonlinear terms cancel out and the canonical generators integrate as

$$Q[\epsilon, \eta] = \frac{k}{8\pi} \int d\phi (\epsilon\mathcal{L} - \eta\mathcal{P}). \tag{2.14}$$

Expanding in modes according to $L_m = \frac{k}{8\pi} \int d\phi e^{im\phi}\mathcal{L}$ and $P_m = \frac{k}{4\pi} \int d\phi e^{im\phi}\mathcal{P}$, the Poisson brackets of the conserved charges reproduce the semidirect sum of a Virasoro and a $u(1)$ Kac–Moody algebra

$$\begin{aligned}[L_m, L_n] &= (m - n)L_{m+n} + \frac{c}{12}m^3\delta_{m+n,0}, \\ [L_m, P_n] &= -nP_{m+n}, \\ [P_m, P_n] &= \frac{k_{\text{KM}}}{2}m\delta_{m+n,0},\end{aligned}\tag{2.15}$$

with $\{, \} \rightarrow i[,]$. In terms of the Newton constant and the AdS_3 radius, the Brown–Henneaux central charge and the Kac–Moody level are given by (1.1). Thus, as naturally expected from different holographic WCFT_2 proposals [35, 42, 48], the Kac–Moody level is also found to be negative.

It is worth highlighting that the asymptotic symmetry generators plainly span the warped conformal algebra in its standard form, without requiring additional mappings of the generators.

3 Modular transformations, asymptotic growth of the number of states and black hole entropy

In the Euclidean continuation, our boundary conditions allow to recover some of the known properties of WCFT s under modular transformations in a straightforward way from purely geometric grounds in the bulk. Indeed, the ground state is clearly identified as the Euclidean AdS_3 geometry. Thus, following the lines of [49, 50] one can prove that Euclidean AdS_3 and BTZ geometries with the boundary conditions in (2.1) turn out to be diffeomorphic provided that the corresponding chemical potentials are related as

$$\mu_{\mathcal{L}\text{AdS}} = \frac{4\pi^2}{\mu_{\mathcal{L}}}, \quad \mu_{\mathcal{P}\text{AdS}} = \frac{2i\pi\mu_{\mathcal{P}}}{\mu_{\mathcal{L}}}. \quad (3.1)$$

This result precisely reproduces the behavior of the temperature and angular potential under an S -modular transformation found in [35], that was directly obtained from a two-dimensional WCFT ⁴. Thus, since the partition function of a WCFT_2 transforms anomalously under (3.1),

$$Z(\mu_{\mathcal{L}}, \mu_{\mathcal{P}}) = e^{\frac{k_{\text{KM}}\mu_{\mathcal{P}}^2}{16\mu_{\mathcal{L}}}} Z\left(\frac{4\pi^2}{\mu_{\mathcal{L}}}, \frac{2i\pi\mu_{\mathcal{P}}}{\mu_{\mathcal{L}}}\right), \quad (3.2)$$

and the spectrum possesses a gap, under the standard mild assumptions one obtains that the asymptotic growth of the number of states is described by

$$S_{\text{WCFT}} = \frac{4i\pi}{k_{\text{KM}}} P_0 P_0^{\text{AdS}} + 4\pi \sqrt{-\left(L_0 - \frac{P_0^2}{k_{\text{KM}}}\right) \left[L_0^{\text{AdS}} - \frac{(P_0^{\text{AdS}})^2}{k_{\text{KM}}}\right]}, \quad (3.3)$$

in full agreement with the “warped Cardy formula” found in [35], where in our case the ground state is precisely identified as the Euclidean AdS_3 geometry.

For Euclidean BTZ and AdS_3 , the zero modes are respectively given by,

$$L_0 = \frac{k}{4}\mathcal{L}, \quad P_0 = \frac{k}{2}\mathcal{P}; \quad L_0^{\text{AdS}} = 0, \quad P_0^{\text{AdS}} = \frac{ik}{2}, \quad (3.4)$$

so that replacing (3.4) and the Kac–Moody level (1.1) into (3.3), one obtains

$$S_{\text{WCFT}} = k\pi \left(\mathcal{P} + \sqrt{\mathcal{L} + \mathcal{P}^2}\right) = \frac{\mathcal{A}}{4G}, \quad (3.5)$$

and hence, the Bekenstein–Hawking entropy is exactly recovered from the warped Cardy formula.

⁴Our chemical potentials relate to theirs according to $\mu_{\mathcal{P}} = 2\beta$ and $\mu_{\mathcal{L}} = i\theta$. Note that the properties under modular transformations naturally agree with that of a chiral CFT with a $u(1)$ current, see e.g., [51].

4 Ending remarks

It is worth mentioning that if one just requires that the variation of the global charges are integrable and finite, the asymptotic behavior of the metric (2.1) could be slightly relaxed. However, consistency of the asymptotic structure implies that the field equations must be fulfilled near the boundary, which inevitably brings one back to the (sub)leading terms of (2.1).

A thorough analysis of the full asymptotic structure including details concerning the canonical generators, modular transformations as well as some aspects about unitarity and parity violation are going to be discussed in a forthcoming publication [52].

Some remarks regarding the unitarity bound, given by

$$L_0 - \frac{P_0^2}{k_{\text{KM}}} \geq -\frac{c}{24}, \quad (4.1)$$

are in order. Indeed, in the context of holographic WCFs, as pointed out in [43], it is useful to assume that the bound still holds in the case of negative k_{KM} , because in spite of the violation of unitarity, the modular bootstrap appears to work very successfully, provided that the primaries possess positive norm, which becomes compatible with states of imaginary $u(1)$ charge.

It is then worth noting that in our context, by means of (3.4), in the case of the black hole, the unitarity bound in (4.1) reduces to

$$\frac{c}{24} (\mathcal{L} + \mathcal{P}^2 + 1) \geq 0, \quad (4.2)$$

which is always fulfilled by virtue of the geometric bounds in (2.8); while the unitarity bound saturates in the case of Euclidean AdS_3 .

Besides, one may naturally wonder about the relationship between our chemical potentials $\mu_{\mathcal{L}}$, $\mu_{\mathcal{P}}$ and global charges $\mathcal{L}$, $\mathcal{P}$, with the standard ones in the canonical ensemble, i.e., with the inverse Hawking temperature β , the angular velocity of the horizon Ω_h , as well as the mass M and angular momentum J . The standard quantities naturally arise in the case of Brown–Henneaux boundary conditions [53] for which the leading terms of the lapse and shift functions, respectively given by $\mathcal{N}$ and $\mathcal{N}^\phi$, are held fixed at infinity. The comparison with our boundary conditions can then be readily performed by extracting $\mathcal{N}$ and $\mathcal{N}^\phi$ from the asymptotic behavior in (2.1). For the sake of this discussion, the generic expression is not so enlightening; nonetheless, in the case of black hole thermodynamics it simplifies as

$$\mathcal{N} = \frac{\mu_{\mathcal{P}} \ell}{4\mathcal{P}}, \quad \mathcal{N}^\phi = -\mu_{\mathcal{L}} + \frac{\mu_{\mathcal{P}}}{4\mathcal{P}}, \quad (4.3)$$

where it must be emphasized that for our boundary conditions what remains fixed are the chemical potentials $\mu_{\mathcal{P}}$ and $\mu_{\mathcal{L}}$. The Hawking temperature and the angular velocity of the horizon are defined through $\mathcal{N} = \beta$ and $\mathcal{N}^\phi = \beta\Omega_h$, and therefore, the relation with our chemical potentials can be readily extracted from (4.3). Consequently, changing the boundary conditions at infinity amounts to change the ensemble, so that the first law reads

$$\delta S = \frac{k}{4} (\mu_{\mathcal{L}} \delta \mathcal{L} + \mu_{\mathcal{P}} \delta \mathcal{P}) = \beta (\delta M + \Omega_h \delta J), \quad (4.4)$$

with

$$M = \frac{k}{4\ell} (\mathcal{L} + 2\mathcal{P}^2) , \quad J = -\frac{k}{4}\mathcal{L} . \quad (4.5)$$

One then obtains that the warped Cardy formula maps to the standard one, since

$$S_{\text{WCFT}} [L_0, P_0] = \frac{\mathcal{A}}{4G} = S_{\text{Cardy}} [L_0^+, L_0^-] , \quad (4.6)$$

which manifestly expresses the Bekenstein–Hawking entropy in terms of the corresponding extensive variables in both ensembles, i.e., in terms of the zero modes of the Virasoro–Kac–Moody algebra in (3.4) and those of two copies of the Virasoro algebra, given by $L_0^\pm = \frac{1}{2}(\ell M \pm J)$, corresponding to the different boundary conditions.

Finally, it is also interesting to compare our boundary conditions with those proposed by CSS in [42]. Their variables are given by chiral functions of $x^+ = \frac{t}{\ell} + \phi$, given by $\bar{L}$ and $\partial_+ \bar{P}$, and an arbitrary constant Δ that is fixed without variation. The global charges manifestly depend on this constant, so that the Kac–Moody level is fixed in terms of the zero mode of the $u(1)$ generators according to $k_{\text{CSS}} = -4\Delta = -4\mathcal{P}_0^{\text{CSS}}$, being positive for AdS_3 ($\Delta = -k/4$) and becomes negative for the black hole ($\Delta > 0$). The Virasoro–Kac–Moody algebra (2.15) is then recovered along the lines of [35] by means of a nonlocal redefinition of the modes, induced by state–dependent asymptotic Killing vectors, so that the level no longer depends on $\mathcal{P}_0^{\text{CSS}}$.

In order to make contact with our boundary conditions, it is necessary to perform a coordinate transformation so that the radial coordinates fit. Chiralities then match provided one of the chemical potentials is chosen as $\mu_{\mathcal{L}} = -1$. The relationship with our chiral functions $\mathcal{L}$ and $\mathcal{P}$, and the remaining fixed chemical potential $\mu_{\mathcal{P}}$ can be seen as follows. By comparing the subleading relevant terms of both spatial metrics one readily finds that

$$\mathcal{L} + \mathcal{P}^2 = \frac{4}{k} \bar{L} , \quad (4.7)$$

while the relation between the remaining variables is recovered by comparing the metrics at the boundary. Both boundary metrics are locally flat and conformally related by a chiral conformal factor according to

$$ds_{b,\text{CSS}}^2 = (1 + \partial_+ \bar{P}) ds_b^2 , \quad (4.8)$$

with ds_b^2 given by (2.7), and

$$\mu = \frac{\partial_+ \bar{P} - 1}{\partial_+ \bar{P} + 1} . \quad (4.9)$$

Since μ is defined through (2.5), by virtue of (2.3), one finds that

$$\mathcal{P} = - \left(\frac{\mu_{\mathcal{P}}}{4} + \partial_+ \ln \right) (1 + \partial_+ \bar{P}) . \quad (4.10)$$

This last equation implies that the CSS boundary conditions can be seen as a subsector of ours, provided that the zero mode of the $u(1)$ current P_0 is fixed without variation, and the

remaining chemical potential is fixed according to

$$\mu_{\mathcal{P}} = -\frac{8}{k}P_0 = -8\sqrt{\frac{\mathcal{P}_0^{\text{CSS}}}{k}} = -8\sqrt{\frac{\Delta}{k}}. \quad (4.11)$$

As a closing remark, following the lines of CSS [42], it is worth exploring the features of the classical WCFT that would emerge from the Hamiltonian reduction of General Relativity with our boundary conditions.

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References

- [1] D.M. Hofman and A. Strominger, *Chiral Scale and Conformal Invariance in 2D Quantum Field Theory*, *Phys. Rev. Lett.* **107** (2011) 161601 [[1107.2917](#)].
- [2] J. Polchinski, *Scale and Conformal Invariance in Quantum Field Theory*, *Nucl. Phys. B* **303** (1988) 226.
- [3] M. Guica, T. Hartman, W. Song and A. Strominger, *The Kerr/CFT Correspondence*, *Phys. Rev. D* **80** (2009) 124008 [[0809.4266](#)].
- [4] A. Castro, D.M. Hofman and G. Sárosi, *Warped Weyl fermion partition functions*, *JHEP* **11** (2015) 129 [[1508.06302](#)].
- [5] K. Jensen, *Locality and anomalies in warped conformal field theory*, *JHEP* **12** (2017) 111 [[1710.11626](#)].
- [6] W. Song and J. Xu, *Correlation Functions of Warped CFT*, *JHEP* **04** (2018) 067 [[1706.07621](#)].
- [7] W. Song and J. Xu, *Structure Constants from Modularity in Warped CFT*, *JHEP* **10** (2019) 211 [[1903.01346](#)].
- [8] H.R. Afshar, *Warped Schwarzian theory*, *JHEP* **02** (2020) 126 [[1908.08089](#)].
- [9] B. Chen, P.-x. Hao and Y.-j. Liu, *Supersymmetric Warped Conformal Field Theory*, *Phys. Rev. D* **102** (2020) 065016 [[2006.04112](#)].
- [10] A. Aggarwal, A. Castro, S. Detournay and B. Mühlmann, *Near-extremal limits of warped CFTs*, *SciPost Phys.* **15** (2023) 056 [[2211.03770](#)].
- [11] R. Poddar, *TT^- deformations of holographic warped CFTs*, *Phys. Rev. D* **108** (2023) 105016 [[2305.15476](#)].
- [12] D. Anninos, J. Samani and E. Shaghoulian, *Warped Entanglement Entropy*, *JHEP* **02** (2014) 118 [[1309.2579](#)].
- [13] A. Castro, D.M. Hofman and N. Iqbal, *Entanglement Entropy in Warped Conformal Field Theories*, *JHEP* **02** (2016) 033 [[1511.00707](#)].

- [14] W. Song, Q. Wen and J. Xu, *Modifications to Holographic Entanglement Entropy in Warped CFT*, *JHEP* **02** (2017) 067 [[1610.00727](#)].
- [15] L. Apolo, S. He, W. Song, J. Xu and J. Zheng, *Entanglement and chaos in warped conformal field theories*, *JHEP* **04** (2019) 009 [[1812.10456](#)].
- [16] B. Chen, P.-X. Hao and W. Song, *Rényi mutual information in holographic warped CFTs*, *JHEP* **10** (2019) 037 [[1904.01876](#)].
- [17] L. Apolo, H. Jiang, W. Song and Y. Zhong, *Swing surfaces and holographic entanglement beyond AdS/CFT*, *JHEP* **12** (2020) 064 [[2006.10740](#)].
- [18] L. Apolo, H. Jiang, W. Song and Y. Zhong, *Modular Hamiltonians in flat holography and (W)AdS/WCFT*, *JHEP* **09** (2020) 033 [[2006.10741](#)].
- [19] S. Detournay, D. Grumiller, M. Riegler and Q. Vandermiers, *Uniformization of Entanglement Entropy in Holographic Warped Conformal Field Theories*, [2006.16167](#).
- [20] B. Chen, Y. Liu and B. Yu, *Reflected entropy in AdS₃/WCFT*, *JHEP* **12** (2022) 008 [[2205.05582](#)].
- [21] S. Detournay and C. Zwikel, *Phase transitions in warped AdS₃ gravity*, *JHEP* **05** (2015) 074 [[1504.00827](#)].
- [22] L. Donnay and G. Giribet, *Holographic entropy of Warped-AdS₃ black holes*, *JHEP* **06** (2015) 099 [[1504.05640](#)].
- [23] H. Afshar, S. Detournay, D. Grumiller and B. Oblak, *Near-Horizon Geometry and Warped Conformal Symmetry*, *JHEP* **03** (2016) 187 [[1512.08233](#)].
- [24] H. Afshar, S. Detournay, D. Grumiller, W. Merbis, A. Perez, D. Tempo et al., *Soft Heisenberg hair on black holes in three dimensions*, *Phys. Rev. D* **93** (2016) 101503 [[1603.04824](#)].
- [25] H. Afshar, D. Grumiller, W. Merbis, A. Perez, D. Tempo and R. Troncoso, *Soft hairy horizons in three spacetime dimensions*, *Phys. Rev. D* **95** (2017) 106005 [[1611.09783](#)].
- [26] A. Castro, C. Keeler and P. Szepietowski, *Tweaking one-loop determinants in AdS₃*, *JHEP* **10** (2017) 070 [[1707.06245](#)].
- [27] A. Aggarwal, A. Castro and S. Detournay, *Warped Symmetries of the Kerr Black Hole*, *JHEP* **01** (2020) 016 [[1909.03137](#)].
- [28] L. Ciambelli, S. Detournay and A. Somerhausen, *New chiral gravity*, *Phys. Rev. D* **102** (2020) 106017 [[2008.06793](#)].
- [29] A. Aggarwal, L. Ciambelli, S. Detournay and A. Somerhausen, *Boundary conditions for warped AdS₃ in quadratic ensemble*, *JHEP* **22** (2020) 013 [[2112.13116](#)].
- [30] A. Aggarwal, A. Castro, S. Detournay and B. Mühlmann, *Near-extremal limits of warped black holes*, *SciPost Phys.* **15** (2023) 083 [[2304.10102](#)].
- [31] J. Xu, *Gravitational radiations of Kerr black hole from warped symmetries*, *JHEP* **04** (2024) 090 [[2310.03532](#)].
- [32] S. Detournay and Q. Vandermiers, *Geometric actions for Lower Spin Gravity*, *JHEP* **10** (2024) 024 [[2408.13198](#)].
- [33] K. Hristov and R.G. Pozzi, *Warped AdS₃ black hole thermodynamics and the charged Cardy formula*, *SciPost Phys. Core* **9** (2026) 039 [[2407.07150](#)].

- [34] S. Detournay, J. Figueroa and A. Vilar López, *Asymptotic T-duality in three dimensions*, *SciPost Phys.* **19** (2025) 044 [[2412.16136](#)].
- [35] S. Detournay, T. Hartman and D.M. Hofman, *Warped Conformal Field Theory*, *Phys. Rev. D* **86** (2012) 124018 [[1210.0539](#)].
- [36] G. Compere and S. Detournay, *Boundary conditions for spacelike and timelike warped AdS_3 spaces in topologically massive gravity*, *JHEP* **08** (2009) 092 [[0906.1243](#)].
- [37] M. Henneaux, C. Martinez and R. Troncoso, *Asymptotically warped anti-de Sitter spacetimes in topologically massive gravity*, *Phys. Rev. D* **84** (2011) 124016 [[1108.2841](#)].
- [38] W. Song and A. Strominger, *Warped AdS_3 /Dipole-CFT Duality*, *JHEP* **05** (2012) 120 [[1109.0544](#)].
- [39] S. Detournay and M. Guica, *Stringy Schrödinger truncations*, *JHEP* **08** (2013) 121 [[1212.6792](#)].
- [40] L. Apolo and M. Porrati, *Free boundary conditions and the AdS_3/CFT_2 correspondence*, *JHEP* **03** (2014) 116 [[1401.1197](#)].
- [41] D.M. Hofman and B. Rollier, *Warped Conformal Field Theory as Lower Spin Gravity*, *Nucl. Phys. B* **897** (2015) 1 [[1411.0672](#)].
- [42] G. Compère, W. Song and A. Strominger, *New Boundary Conditions for AdS_3* , *JHEP* **05** (2013) 152 [[1303.2662](#)].
- [43] L. Apolo and W. Song, *Bootstrapping holographic warped CFTs or: how I learned to stop worrying and tolerate negative norms*, *JHEP* **07** (2018) 112 [[1804.10525](#)].
- [44] M. Henneaux, A. Perez, D. Tempo and R. Troncoso, *Chemical potentials in three-dimensional higher spin anti-de Sitter gravity*, *JHEP* **12** (2013) 048 [[1309.4362](#)].
- [45] C. Bunster, M. Henneaux, A. Perez, D. Tempo and R. Troncoso, *Generalized Black Holes in Three-dimensional Spacetime*, *JHEP* **05** (2014) 031 [[1404.3305](#)].
- [46] A. Pérez, D. Tempo and R. Troncoso, *Boundary conditions for General Relativity on AdS_3 and the KdV hierarchy*, *JHEP* **06** (2016) 103 [[1605.04490](#)].
- [47] T. Regge and C. Teitelboim, *Role of Surface Integrals in the Hamiltonian Formulation of General Relativity*, *Annals Phys.* **88** (1974) 286.
- [48] T. Azeyanagi, S. Detournay and M. Riegler, *Warped Black Holes in Lower-Spin Gravity*, *Phys. Rev. D* **99** (2019) 026013 [[1801.07263](#)].
- [49] S. Carlip and C. Teitelboim, *Aspects of black hole quantum mechanics and thermodynamics in (2+1)-dimensions*, *Phys. Rev. D* **51** (1995) 622 [[gr-qc/9405070](#)].
- [50] J.M. Maldacena and A. Strominger, *$AdS(3)$ black holes and a stringy exclusion principle*, *JHEP* **12** (1998) 005 [[hep-th/9804085](#)].
- [51] P. Kraus and F. Larsen, *Partition functions and elliptic genera from supergravity*, *JHEP* **01** (2007) 002 [[hep-th/0607138](#)].
- [52] K. Lara and R. Troncoso. *Work in progress*.
- [53] J.D. Brown and M. Henneaux, *Central Charges in the Canonical Realization of Asymptotic Symmetries: An Example from Three-Dimensional Gravity*, *Commun. Math. Phys.* **104** (1986) 207.